

Harith Al-Safi

📍 Leeds, UK 📞 +447444585915 @ harith.alsafi@gmail.com 🌐 https://harith.io/

Electronics and Computer Engineer | C++ Engineer

🐙 github.com/harith-alsafi 🌐 linkedin.com/in/harith-al-safi

📄 About me

An experienced Electronic & Computer Engineer specializing in full-stack development and AI/ML applications, with a proven track record in developing high-performance systems and leading technical teams.

📁 Experience

Fullstack Engineer

📅 Sep 2024 - Ongoing

BT Group

📍 Manchester, UK

- Architected high-performance REST APIs using Java Spring Boot with MongoDB achieving 99.9 % uptime
- Implemented deployment architecture using Pulumi, Kafka and AWS improving response time by 60 %
- Established mentorship program for apprenticeship students, resulting in 90 % pass rate for training assessments
- Led regional Grad network initiatives, organizing monthly tech talks and workshops with 50+ regular attendees

Frontend Developer

📅 Aug 2023 - Nov 2023 (4 Months)

Velz Travel

📍 Remote

- Directed the frontend of a travel web app for a startup built with Next.js, React.js and Typescript
- Developed user interface using Material UI following Figma design guidelines and business logic using Node.js
- Collaborated with developers using Jira and Git whilst working on Java Springboot with PostgreSQL and Docker
- Achieved 93% test coverage through comprehensive unit testing, resulting in 70% reduction in production bugs

Software Engineer (Placement Year)

📅 July 2022 - July 2023 (12 Months)

Johnson Controls

📍 London UK

- Administered C \C++ projects while fixing bugs, adding features, refactoring and migrating 90% of codebase to C#
- Developed C# .NET libraries for fire detection simulations and data analysis improving operations by 32%
- Collaborated on an analytics web dashboard with React.js for predictive maintenance with dev ops and CI/CD
- Invented a custom scripting language with a compiler for C# app increasing development productivity by 40%
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure
- Managed a team of 10 interns to create a business plan and ended up placing 2nd amongst the whole company

C++ Developer Freelance

📅 Dec 2021 - Mar 2022 (4 Months)

Paperound

📍 Leeds UK

- Developed signal processing filters and modulators using SciPy Python, C++ and Fourier analysis for RF signals
- Developed distributed system using C++ OpenMP, OpenCL and CUDA for parallel statistical computing
- Developed low level ARM embedded systems with C whilst using memory allocation and register programming
- Built a lexer, compiler and assembler with C for an 8-bit CPU built with Logic Gates and Verilog
- Created a desktop application using C++ and QT for video analysis and processing
- Engineered a high performance C++ linear algebra library for data science through Jenkins, Docker and Cmake

IT administrator

📅 Jun 2020 - Mar 2021 (10 Months)

Mesopotamia Group

📍 Amman Jordan

- Collaborated with a team to build a local file server on Ubuntu built using Apache and Nextcloud
- Administered the server using Bash, Python and Nmap with SSH tunneling and port forwarding
- Engineered a load balancer using MPI and C to distribute traffic between servers increasing availability by 34%
- Analyzed the network data logs using Python to predict and prevent cyber attacks
- Implemented Linux networking and protocols using IP addressing and routing for cloud hosting

📖 Education

BEng (Hons) in Electronics and Computer

📅 Sept 2019 - June 2024

University of Leeds (UoL)

🏆 First-Class Honors

- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design
- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Research

Skills and Languages

Technical: C \ C++ Python Java C# JavaScript Linux Docker Azure AWS Jira Git
Jenkins Terraforms Kubernetes SQLite MySQL MongoDB PostgreSQL .NET WPF JavaFx
React Js QT Framework Flutter Flask Springboot ASP.NET NumPy KerasRL Matplotlib
Pytorch SciPy TensorFlow Pandas OpenCv CMake Node.js OpenCL OpenGL OpenMP
CUDA Arm Mbed Os Arm Keil FPGA Verilog HDL Arduino Blender Photoshop AutoCAD

Soft: Management Leadership Motivated Deterministic Communication Problem solver

Languages: Arabic (Full Professional) English (Full Professional) Spanish (Elementary)

Projects

MCPP: High Performance C++ Library for Numerical Computation  Apr 2021 - Oct 2022  Personal

- Created a cross-platform math library for C++ used for matrix algebra, statistics and machine learning.
- Engineered with CI/CD using Jenkins, unit tests, Docker, Cmake, Doxygen and Git for version control
- Improved performance by 40% using parallel computing with OpenMP, OpenCL and CUDA

TBot: Machine Learning Stocks Trading Bot  Apr 2021 - Oct 2021  Personal

- Developed gradient boosting machine learning model using Python and Scikit-learn for stock prediction
- Anlayzed and processed data using Pandas, Numpy and Scipy for training and testing the model
- Backtested the model using historical data and simulated trading using Python and Matplotlib
- Developed a REST API and a user interface to interact with the bot using AWS, Flask and PyQT

Vega: LLM Chat Web App for IoT Circuits  Oct 2023 - Jun 2024  UoL

- Developed a full stack chat bot web application with Typescript, React.js, Next.js, TailwindCSS and NoSQL
- Programmed natural language processing algorithms using LLMs to interpet and execute user messages
- Developed a Python Flask REST Server with SQLite on the Raspberry Pi to communicate with the chat bot
- Engineered a circuit with sensors, fans, LCD and buttons to execute the user's commands from the chat bot

Professional Courses

Data Analysis with Python  Oct 2021

- Develop Python code for cleaning and preparing data for analysis
- Perform exploratory data analysis and analytical techniques to datasets using Pandas, Numpy and Scipy
- Manipulate data using dataframes, understand data distribution, perform correlation and create data pipelines
- Build and evaluate regression models using machine learning scikit-learn for prediction and decision making

Databases and SQL for Data Science with Python  Sept 2021

- Analyze data within a relational database on the cloud using IBM DB2, SQL, Python
- Construct SQL CRUD operations and compose views, transactions, stored procedures and joins.

Python for Data Science, AI & Development  Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, Numpy, and Beautiful Soup
- Access web data using APIs and web scraping from Python in Jupyter Notebooks.

Extracurricular

AI Agent hackathon  Mar 2023
Lablab  Remote

Web Developer hackathon  Feb 2023
Royal Hackaway v6  London, UK

Event Coordinator Volunteer  Jan - 2023
IEE RAS ICRA  London, UK